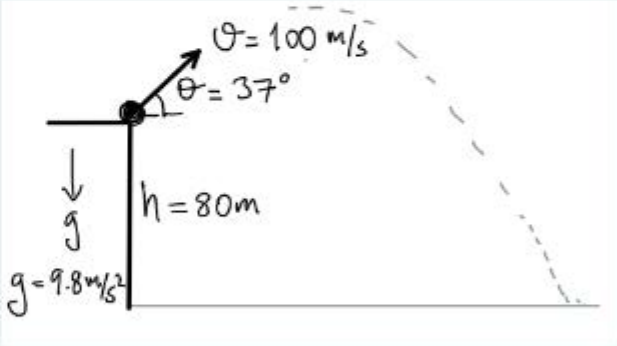
Physics I : FAST and SHORT EXAM

Points: 35/100

✓ **Correct** 5/5 Points

1

A ball is located 80m above the sea level. Its initial velocity is 100 m/s and the angle between the horizontal is +37 degrees. What is the magnitude of the final velocity of the ball just before it hits the surface?

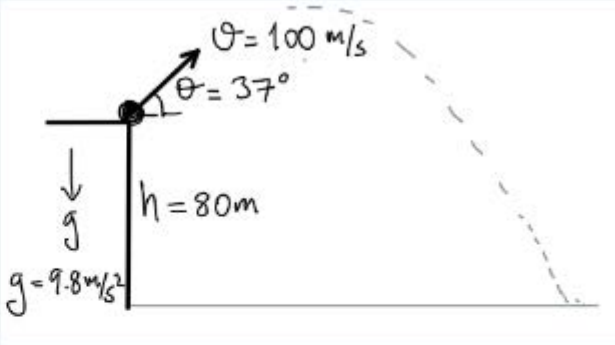


- (●) $107.8 \frac{m}{s}$ ✓
- ( ) $80 \frac{m}{s}$
- ( ) $113.1 \frac{m}{s}$
- ( ) $100 \frac{m}{s}$

✕ **Incorrect** 0/5 Points

2

1. A ball is located 80m above the sea level. Its initial velocity is 100 m/s and the angle between the horizontal is +37 degrees. How long will it take for the ball to hit the surface?

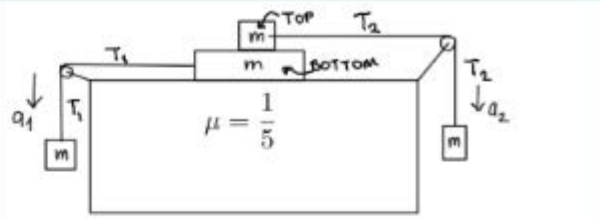


- 11.9 s ✓
- 13.5 s
- 16 s
- 12 s
- 71.9 s

✓ **Correct** 5/5 Points

3

What is total frictional force acting on the TOP mass ?



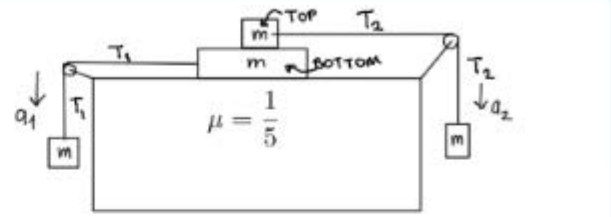
There are four objects with the same mass in the setup as given in the figure . On horizontal surface of the table, the mass are top of each other. Top mass is attached to the mass m on the right and they move with acceleration a_2 and the bottom mass m is attached to the mass m on the left and they move with an acceleration of a_1. All surfaces have the same friction coefficient of $\mu = \frac{1}{5}$.

- mg
- $mg/5$ ✓
- $4mg/5$
- $2mg/5$
- $3mg/5$

✓ **Correct** 5/5 Points

4

What is the total frictional force acting on the BOTTOM mass ?



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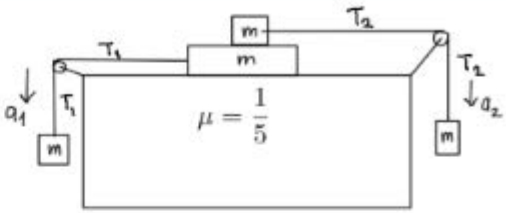
- mg
- $mg/5$
- $4mg/5$
- $3mg/5$ ✓

✕ **Incorrect** 0/5 Points

5

What is a1 acceleration?



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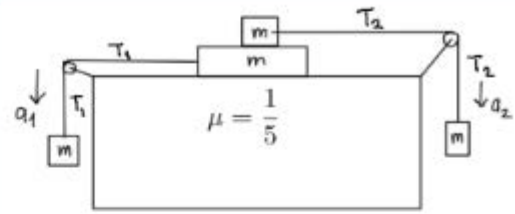
- g
- $g/2$
- $g/5$ ✓
- $2g/5$
- $3g/5$
- $4g/5$

✓ **Correct** 5/5 Points

6

What is a2 acceleration?



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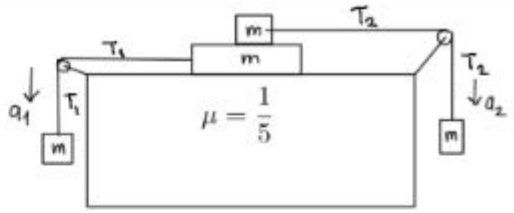
- g
- $g/2$
- $g/5$
- $2g/5$ ✓
- $3g/5$
- $4g/5$

✕ **Incorrect** 0/5 Points

7

What is T1 tension (the tension of the left rope?)



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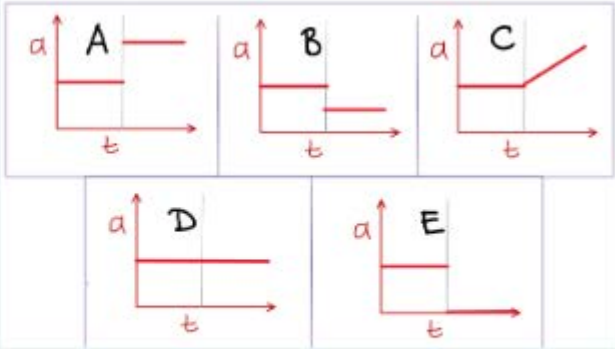
- mg
- $mg/2$
- $4mg/5$ ✓
- $2mg/5$
- $3mg/5$

8

What will be the a1 graphics after the top mass falls off ? (Yukaridaki kucuk kutle dusunce a1 ivmesinin zamana gore grafigi nasildir?) Grafiklerde dusmeden once ve sonra zamani gri cizgiyle belirtilmistir.

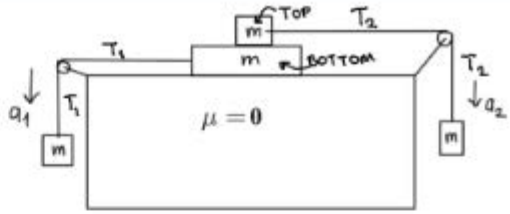


- (selected) *A* ✓
- *B*
- C
- D
- E

9

The same setup is given as shown in the figure but now the frictional coefficient is ZERO, what would be a1 and a2 ?

a1 ::: a2



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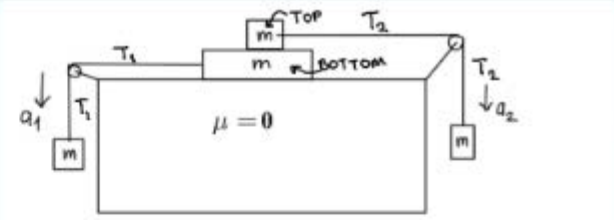
- g ::: g
- $\frac{g}{2}$ ::: $\frac{g}{2}$ ✓
- $\frac{2g}{5}$ ::: $\frac{g}{5}$
- $\frac{2g}{5}$ ::: $\frac{3g}{5}$
- $\frac{2g}{3}$ ::: $\frac{2g}{5}$

✓ **Correct** 5/5 Points

10

The same setup is given as shown in the figure but now the frictional coefficient is ZERO, what would be a1 and a2 ?

T1 ::: T2



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- mg ::: mg
- $\frac{g}{2}$ ::: $\frac{g}{2}$ ✓
- $\frac{2mg}{5}$ ::: $\frac{mg}{5}$
- $\frac{2mg}{3}$ ::: $\frac{mg}{2}$
- $\frac{2mg}{3}$ ::: $\frac{mg}{5}$
- $\frac{3mg}{4}$ ::: $\frac{mg}{4}$

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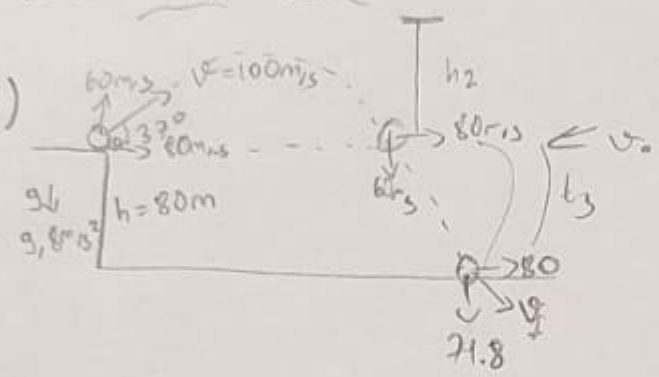
t_1 t_2

1)



$V_f^2 = V_i^2 + 2\Delta x\, a$

$V_f^2 = 10^4 + 2 \cdot 80 \cdot 9{,}8$

$v_f = 107.55$

2) $0 = 3600 + 19.6 h_2$

$h_2 = 183.6$

$= 4.9t^2$

$t_1 = 6.1$

$t_2 = 6.1$

$x = vt$

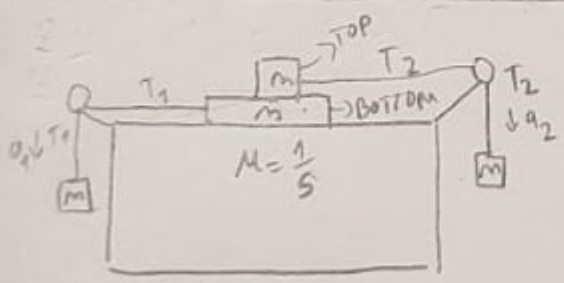
$80 = 65{,}9 \cdot t_3$

$t_3 = 1.2$

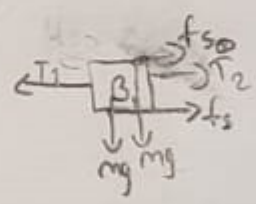
$t_1 + t_2 + t_3$

$= 13.4$

$\underline{13.5}$



3)



$mg - T_1 = m \cdot a_1$

$mg - T_2 = m \cdot a_2$

TOP:

$T_2 - \frac{1}{5} \cdot mg = m \cdot a_2$

$T_2 = m\left(\frac{g}{5} + a_2\right)$

$g - a_2 = \frac{g}{5} + a_2$

$2a_2 = \frac{4g}{5}$

$\boxed{a_2 = \frac{2g}{5}}$

$T_1 = m(g - a_1)$

$T_2 = m(g - a_2)$

$\boxed{f_s = \frac{2mg}{5}}$ $\boxed{f_{s_0} = \frac{mg}{5}}$

$T_1 - T_2 - f_{s_0} - f_s = m \cdot a_1$

$\cancel{m}(g - a_1) - \cancel{m}\left(\frac{3g}{5}\right) - \frac{3mg}{5} = \cancel{m} \cdot a_1$

$g - a_1 = a_1$

$\boxed{a_1 = \frac{g}{2}}$

3) $f_{s_0} = \frac{mg}{5}$

4) $f_{s_0} + f_s = \frac{3mg}{5}$

5) $a_1 = \frac{g}{2}$

6) $a_2 = \frac{2g}{5}$

7) $T_1 = m \cdot \left(\cancel{g - \frac{g}{2}}\right) = \frac{mg}{2}$

8) after falling

~~$T_1 \downarrow a_1 \uparrow$~~

same as before

$mg - T_1 = m \cdot a_1$

$T_1 = mg - m \cdot a_1$

$T_1 = m \cdot a_1$

$\cancel{m}g - \cancel{m}a_1 = \cancel{m}a_1$

$g = 2a_1$

$\boxed{a_1 = \frac{g}{2}}$

$mg - T_2 = m \cdot a_2$

$T_2 = mg - m \cdot a_2$

$T_2 = m \cdot a_2$

$\cancel{m}g = 2\cancel{m}a_2$

$g = 2a_2$

$\boxed{a_2 = \frac{g}{2}}$

$T_1 = \frac{mg}{2}$ $T_2 = \frac{mg}{2}$

Question 9 is $\frac{g}{2}$ $\frac{g}{2}$

Question 10 is